

SUBJECT: Scope of Work for 1030 nm Femtosecond Fiber Laser System

1. Objective

The vendor shall supply one (1) industrial-grade 1030 nm femtosecond fiber laser system suitable for precision micromachining and nonlinear material processing research at Brookhaven National Laboratory (BNL).

2. Scope of Supply

The vendor shall provide a complete, fully integrated laser system including:

- One (1) Ytterbium (Yb)-doped femtosecond fiber laser (or equivalent industrial femtosecond platform) operating near 1030 nm
- Rack-mountable control unit
- Separate compact optical head
- Interconnecting fiber/umbilical cable (minimum 1.0 meter in length; longer length preferred)
- All required power supplies, control electronics, safety interlocks, and cabling
- User manual and system documentation

The system shall be fully assembled, factory tested, and ready for operation upon installation at BNL.

3. Technical Requirements

The laser system shall meet or exceed the following minimum performance requirements:

Optical Characteristics

- Center wavelength: 1030 nm $\pm$ 10 nm
- Pulse duration: < 500 fs
- Pulse energy: > 40 μ J at approximately 500 kHz repetition rate
- Average output power: > 20 W
- Repetition rate: Adjustable from 50 kHz to 500 kHz, with capability up to 2 MHz at reduced pulse energy
- Beam quality (M^2): < 1.4
- Linear polarization with extinction ratio > 20 dB
- Free-space or collimated output suitable for integration into a micromachining station

Stability and Performance

- Long-term power stability: < 1% RMS over 8 hours
- Pulse-to-pulse stability: < 1% RMS
- Beam pointing stability: < 50 μ rad
- Warm-up time: < 15 minutes from cold start to full specification

Mechanical and Form Factor

- Rack-mountable control electronics
- Compact optical head with standard mechanical mounting features
- Umbilical connection between control unit and optical head of at least 1.0 meter

Electrical and Control Interfaces

- Universal single-phase AC input: 100–240 VAC, 50/60 Hz
- Remote control capability via RS-232, Ethernet, or USB
- TTL input for on/off gating with bandwidth > 500 kHz
- Analog (0–10 V) or software control for pump diode current or output power modulation
- Synchronized output signal for integration with external motion stages
- Class-4 laser system with appropriate interlock and safety features

Facility and Cooling Requirements

- Air-cooled or water-cooled configuration
- If water-cooled, compatibility with standard laboratory water systems (approximately 20–25 °C); vendor shall specify required flow rate and pressure
- Operation in standard laboratory environment (15–30 °C; 10–80% non-condensing humidity)

4. Documentation and Acceptance

The vendor shall provide a factory test report with the delivered system including:

- Autocorrelation trace verifying pulse duration
- Beam quality (M²) measurement
- Emission spectrum and center wavelength verification
- Long-term power stability data over a minimum of 8 hours

Acceptance will be based on review of documentation and confirmation that the delivered system conforms to the stated technical requirements.

5. Warranty

The vendor shall provide a minimum one (1) year warranty covering parts and labor.

6. NRTL Certified

All standalone devices must be NRTL certified.

7. Delivery

The system shall be delivered to Brookhaven National Laboratory, Upton, NY.

Anticipated delivery of 12–14 weeks after receipt of order.

Signature:  (date) : 02/25/2026

Of Requestor or Technical Rep

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